Unit
Plate Tectonics

Title
5. Seafloor Spreading

Summary
Students take what they know about earthquake, volcano and mid-ocean ridge distributions (The Big One and Plate Patterns) and put it together with what they know about convection in the Earth's mantle (Journey Through Earth and Convection in a Pan). They revisit what they know about how earthquakes are created, by the sudden release of energy as plates collide or rub together (but not so much when they split apart). They look for patterns in their world maps, observing that mid-ocean ridges and dense earthquake/volcano zones tend to lie on the opposite side of plates. With this information, they can infer the direction that the plates are moving. Next students build a model illustrating seafloor spreading and discuss the magnetic and seafloor age data that support this model. Finally, students codify the different types of plate boundaries, describing the various features and characteristics of each.

Objectives
Can find patterns on a map and use that information to infer the direction of plate motion.
Can diagram and explain what causes earthquakes and volcanoes.
Can build and interpret a physical model of seafloor spreading.
Can describe some of the evidence supporting seafloor spreading.
Can describe the characteristics of the 3 major classes of plate boundaries as well as differences between subducting convergent boundaries and continent-continent convergent boundaries.

Vocabulary
Mantle
Lithosphere
Crust
Earthquake
Volcano
Mid-ocean ridge
Tectonic plate
Oceanic plate
Continental plate
Seafloor spreading
Convergent boundary
Divergent boundary
Transform boundary
Subduction
Island arc

Time
30-50 min hypothesize plate motions
20-25 min build seafloor spreading models and discuss evidence

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30-35 min study models and describe types of plate boundaries

Grouping
Inferring the direction of plate motion will occur in groups of 2 or 3. Making models of seafloor spreading takes place individually.

Materials
- Cardstock paper
- Copy of Seafloor Spreading Model Pieces on cardstock for each student
- Colored pencils
- Stapler
- Scissors
- Students' color coded World Earthquake Maps from Plate Patterns lesson
- Large labeled world map from Plate Patterns lesson or a copy of the "This Dynamic Planet" poster from the USGS, $14 + $5 handling http://www.minerals.si.edu/tdpmap/order.htm
- Optional: Pictures, animations or other information showing seafloor spreading, magnetic survey maps of the ocean floor, or age maps of the ocean floor (see sources section for resources).

Setting
Classroom

Teacher Background
Seafloor spreading is one of the most critical pieces of evidence in the development and support of the theory of plate tectonics. It is through this process that new oceanic crust is formed along the mid-ocean ridges as oceanic plates diverge and separate. Magma wells up into the gap, hardens, and forms new crust. As the plates continue to separate, these newly formed pieces of oceanic crust separate and make room, gradually moving outward away from the mid-ocean ridges at the rate of a few centimeters per year.

After several million years of this slow journey away from the mid-ocean ridges, the oceanic crust collides with a different tectonic plate in a process known as subduction. If the oceanic crust meets continental crust, the denser oceanic crust is forced under the continental crust. If the oceanic crust meets oceanic crust, then one or the other will be forced below the other. In these subduction zones, earthquakes are common due to the build up and sudden release of energy at the junction. Also, as the oceanic crust dives down below the other plate into the mantle, the old plate melts, forming a pool of magma that is forced back up through the crust above as volcanoes on the continent or a chain of volcanic islands on one side of the subduction zone.

In some instances, some continental crust is riding along on the oceanic plate and collides into another piece of continental crust. This is currently taking place in the Himalayas and was part of the process that created California (see the background section of Making California for more details). At these continent-continent convergent boundaries, the two sections of continental crust ram into one another, causing the crust above to buckle and fold into tall mountain ranges (the Himalayas and the Sierra Nevadas).

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Using these patterns, students can infer both the direction and the relative speed each tectonic plate is moving. The following table shows all the plates and their approximate direction and speed.

Tectonic Plate	Approximate direction	Approximate velocity in cm/year
African	W	1.622
Antarctic	SW	2.21
Arabian	NW	2.236
Australian	N	5.808
Caribbean	W	3.03
Cocos	NE	11.571
Eurasian	W	2.252
Indian	N	2.593
Juan de Fuca	NE	10.446
Nazca	E	3.586
North American	W	4.114
Pacific	NW	11.78
Phillippine	NW	10.574
Scotia	W	4.893
South American	W	4.83

For further information on the types of plate boundaries, see the background section of the Plate Patterns lesson.

The ultimate cause of sea floor spreading (and the theory of plate tectonics as a whole) is still debated in the scientific community. Some argue that it is driven by convection currents in the mantle. Others argue that the upwelling of magma and creation of new crust at mid-ocean ridges pushes the older crust out of the way (the “Ridge Push” theory). Others argue that the sinking of the old crust at subduction zones drags the oceanic crust along behind it (the “Slab Pull” theory).

Student Prerequisites

Students should have created or studied a map with data about earthquake, volcano and mid-ocean ridge locations. Students should know that the Earth’s crust is divided into large plates the size of continents or oceans. Students should know about convection currents in the Earth’s mantle and should understand how those could affect the motion of the tectonic plates above.

Getting Ready

1. Make a copy of the Seafloor Spreading Model Pieces on cardstock paper for each student
2. Set out scissors and colored pencils
3. Remind students to bring their color coded World Earthquake Maps from Plate Patterns lesson
4. At the front of the room display the large labeled world map from Plate Patterns lesson or a copy of the “This Dynamic Planet” poster from the USGS

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Lesson Plan

Infer plate motion

1. Review convection currents in the Earth's mantle. Discuss how convection currents form and how they move. (The heat from the core causes mantle in some areas to rise while cooled mantle near the crust sinks.)
2. Discuss how things floating on top of the surface of this mantle move. (At the surface, stuff just above the places where hot mantle rises gets pushed away from each other and stuff where the cool mantle sinks gets pulled down or pushed against the edge of something solid.)
3. Tell students you are going to switch gears briefly.
4. Review the causes of earthquakes. Discuss how earthquakes are caused when two plates collide or rub against one another. For a while, the plates get stuck even though the pressure is building up all the time and finally, when the plates release, you get an earthquake.
5. Discuss how speed might affect the number of earthquakes. Would you expect more earthquakes each year if 2 plates collide at high speed or at low speed?
6. Discuss whether earthquakes would be expected where plates are separating. (Not so much.)
7. Put it together. So, the Earth's surface is made up of plates riding along on convection currents in the mantle. The plates are moving in all different directions. In some places the will collide head on. In some places they will rub against one another going in opposite directions. Where they collide or rub, you'll get lots of earthquakes. In some places they will separate from one another. Where the separate, you'll get much fewer earthquakes.
8. Ask students to get out their color coded World Earthquake Maps.
9. Tell them that their job, in teams of 2 or 3, is to see if they can figure out which direction each plate is moving using the information on their maps (earthquake epicenter locations, volcano zones, mid-ocean ridge locations). Their job is to draw an arrow ***in pencil*** on each plate showing the direction that plate is moving. If they believe a plate is standing still, then don't draw anything at all. If you want to give a hint, tell them to start with the Pacific plate or the Nazca plate and go from there.
10. Give students 10-20 min to discuss their maps and make hypotheses. Not that the Eurasian and African Plates don't fit the pattern as well because they
11. Different groups will figure out the patterns at different rates. For those groups that finish early, have them modify the length of their arrows to show the speed of the plates. For instance, a fast moving plate should have a very long arrow while a slow moving plate should have a short arrow.
12. When most groups have finished, bring the focus of the class up to the large map at the front of the classroom. One by one, go through the plates (starting with the Pacific and Nazca plates) discussing what direction they inferred the plate would move and why. Students should correct their personal maps if there are any errors.
13. Optional: when all the direction information has been added, see if students can figure out how fast each plate is moving. Use the length of the arrow to represent speed.

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Seafloor spreading model

1. On the students' labeled World Earthquake Maps have them focus on a diagonal rectangular strip starting at the South American plate, crossing west over the Nazca and Pacific Plates, past Hawaii, and ending the in earthquake/volcano zone on the right edge of the Phillippine Plate and Korea.
2. Tell students that they will be building a model of this region today.
3. Pass out the Seafloor Spreading Model pieces, the scissors, staplers, and colored pencils.
4. Begin with the page with the diagram cut-through view of the mantle showing convection currents. Fold the paper lengthwise along the fold line "hot dog style". Focus on the large diagram for now.
5. Label the diagram (mantle, mid-ocean ridge, lithosphere, oceanic crust, continental crust, island arc, volcanoes, ocean, convection currents, etc.)
6. Color the diagram in.
    - Use blue to represent the ocean.
    - Use one color to represent the left-most oceanic crust with the volcanic island arc.
    - Use another color to represent the two slabs of oceanic crust in the middle spreading outward from the mid-ocean ridge.
    - Use another color to represent the continental crust on the far right.
    - Use a bright color like orange or red to represent that magma in the mantle and welling up through the volcanoes.
7. Now describe that the big diagram is the side view while the other part on the other side of the fold is the top view. Drape the model over the edge of the table with the large diagram hanging over the side to get everyone oriented.
8. Next label and color the top of the model so that they match the colors on the cut-through view.
9. Use the scissors to cut a slit along each of the dark black lines on the top of the model. ***Do not cut all the way to the edge.*** It helps to make a light fold perpendicular to the line and make a small snip with the scissors to get the cut started. Then you can open the paper up to cut the rest of the line.
10. Now take the other piece of the model (the piece of paper with all the grey stripes). Cut out the grid along the outer lines and throw the edges away.
11. Cut the two strips apart lengthwise down the middle "hot dog style".
12. Put the two strips together, one on top of the other with the writing on the inside, facing one another.
13. Glue or staple the two strips at the end that says "Glue this end to other strip".
14. Feed the stapled/glued end into the slit labeled "mid-ocean ridge".
15. Feed the free end of the left hand strip into the subduction zone on the left.
16. Feed the free end of the right hand strip into the subduction zone on the right.
17. Hold a free end in each hand and slowly pull the strips of paper through the subduction zones. "New" ocean curst should appear at the mid-ocean ridge. Pull down at the mid-ocean ridge to reset the model.
18. Point out the divergent and convergent boundaries on the model.
19. Once the models have been built, discuss what the white/grey stripes represent – magnetic reversals recorded in the rock on the ocean floor in the form of tiny pieces of iron, aligned with the Earth's magnetic field.

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20. Discuss what the age markers mean – the ocean floor nearest the mid-ocean ridges are youngest and the ones nearest the subduction zones are oldest.
21. Point out the mirror-image symmetry on each side of the mid-ocean ridge and how that provided the first strong evidence to scientists that seafloor spreading actually takes place. (Now GPS (global positioning systems) can measure the creep of the plates in millimeters per year, corroborating the evidence trapped in rocks on the ocean floor.)
22. Refer back to the labeled world Earthquake map and the Pacific/Nazca plate boundary area, pointing out the relationship between the map and the model. Notice how subduction zones tend to have lots of earthquakes and volcanoes. If the subduction zone is in the ocean (like the Alaskan Aleutian Islands, the Philippines, the South Pacific Islands, and the Caribbean Islands), a chain of volcanic islands forms (a volcanic island arc). If the subduction zone is on the edge of a continent (like Central America and Southe America), a volcanic mountain range is pushed up on the continent.
23. Finally, discuss the 2 plate boundaries that are not shown in this model: transform boundaries and continent-continent convergent boundaries. Use 2 pieces of paper to have each student model these boundaries.
    - For a transform boundary, slide the 2 pieces of paper past each other, one moving up, and one moving down. Explain how this is similar to what is happening at the edge of California along the San Andreas fault. Notice on the World Earthquake map How transform boundaries thend to have lots of earthquakes but few volcanoes.
    - For a continent-continent convergent boundary, slide the two pieces of paper into one another on the table top but don't let one piece slide below the other like subduction. Instead, the 2 pieces of paper should collide and then buckle, each piece wrinkling up off the table. Explain how this is similar to what is happening at the top of the Indo-Australian plate where India has rammed into the Eurasian Plate. Notice on the World Earthquake Map how continent-continent convergent boundaries have lots of earthquakes and huge mountain ranges.

Assessment

1. Ask students to answer the following questions for each of the 4 types of plate boundaries described in this lesson:
    a. Describe how the plates are moving.
    b. Give an example of somewhere in the world where you can find this kind of boundary.
    c. Are there earthquakes?
    d. Are there volcanoes?
    e. Is there a mid-ocean ridge?
    f. Draw a labeled diagram of this type of boundary from a side view (like the large diagram on the model) showing the plates, the mantle below, and arrows to show what direction the plates are moving.
2. For each of the following places in the world, name the type of boundary that the place is located on. If the place is not on a plate boundary, name the tectonic plate the place is located.
    a. Nepal is a country high in the Himalayas, near the northern border of India. Nepal's landscape is known for dramatic, tall mountain ranges and beautiful secluded valleys. There are many earthquakes.

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b. Martinique is a tropical paradise in the Eastern Caribbean with lush rainforests and white sand beaches. The island is dominated by two currently dormant volcanoes, Mount Pelee and Carbet.
c. The Red Sea is a long, narrow inlet off the Indian Ocean, between Africa and Saudi Arabia. Currently, at its widest point, the sea is 190 miles across but it is slowing getting wider by a few centimeters each year.
d. Where you live.

Going Further

1. Study other evidence for plate tectonics besides sea-floor spreading such as the fit of the continents and fossil evidence. See the Evidence for Plate Tectonics lesson.
2. Study the amazing life forms found at mid-ocean ridges. See the sources section of the Plate Patterns lesson for detailed resources.
3. Createa travel brochure of the geological features and geologic history of some of the worlds most famous plate boundaries. See the Plate Boundaries Around the World lesson for more details.

Sources

This activity combined the ideas from 1) a simple paper Seafloor spreading model by Ellen Metzger (http://www.ucmp.berkeley.edu/fosrec/Metzger3.html) and 2) a more detailed shoebox Seafloor spreading model by John Lahr (http://pubs.usgs.gov/of/1999/ofr-99-0132/). Both lesson plans provide excellent background information and excellent alternative models.

The best resources for more information on Seafloor spreading can be found at:

- This Dynamic Earth has an excellent section on how the theory of plate tectonics was developed using ocean floor data (http://pubs.usgs.gov/gip/dynamic/developing.html#anchor10912731).
- The Wikipedia articles on seafloor spreading (http://en.wikipedia.org/wiki/Seafloor_spreading) and subduction (http://en.wikipedia.org/wiki/Subduction) provide excellent, concise summaries as well as beautiful images.
- The animations on the front page of this lesson were created by the USGS and the National Parks Service (http://www2.nature.nps.gov/geology/usgsnps/animate/pltecan.html).

The table of plate motion was created using the Rice University Plate Motion Calculator (http://terra.rice.edu/department/faculty/gordon/hs3-nuvel1a.html).

Standards

Grade 6

Plate Tectonics and Earth's Structure

1. Plate tectonics accounts for important features of Earth's surface and major geologic events. As a basis for understanding this concept:

a. Students know evidence of plate tectonics is derived from the fit of the continents; the location of earthquakes, volcanoes, and midocean ridges; and the distribution of fossils, rock types, and ancient climatic zones.

b. Students know Earth is composed of several layers: a cold, brittle lithosphere; a hot, convecting mantle; and a dense, metallic core.

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c. Students know lithospheric plates the size of continents and oceans move at rates of centimeters per year in response to movements in the mantle.
d. Students know that earthquakes are sudden motions along breaks in the crust called faults and that volcanoes and fissures are locations where magma reaches the surface.
e. Students know major geologic events, such as earthquakes, volcanic eruptions, and mountain building, result from plate motions.
f. Students know how to explain major features of California geology (including mountains, faults, volcanoes) in terms of plate tectonics.

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